



# RTL Design Sherpa

## STREAM Characterization Monbus Compression Report

June 2026

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# Readme

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## Monbus Compression — Characterization (in-core PMU build)

Companion to `../perf/`. Where the perf report measures how hard the DMA drives the bus, this measures how well the **monbus compressor** + **half-beat packer** shrink the monitor-event stream on real captured traffic.

**Provenance.** The compressor RTL and packet format are unchanged by the RFC Stage E in-core perf-monitor work / arm-gap fix, so the locked 682-record golden capture and its reductions (current **63.0%**, half-beat **75.7%**) still stand. Re-confirmed live on the 2026-06-21 in-core-PMU build (board 210292B7D46F, 8desc\_1ch\_1MB --mon-config debug-compl): CRC ok, decode OK, **63.5%** reduction — consistent with the golden capture.

**Bitstream:** all-fixes build (USE\_MON\_COMPRESSION = USE\_MON\_HALFBEAT = USE\_MON\_CAM\_PIPELINE = 1), WNS +0.048 ns @ 100 MHz, xc7a100t. **Primary data:** the committed encoder CAM logic replayed **bit-exact and offline** on the real 682-record on-board capture (`../compression_dataset/desc_axi_16desc_8ch_1MB.json`). The on-chip packer is verified bit-identical to this model in cosim (`val/amba/test_monbus_halfbeat.py`), so the offline reduction is what the FPGA emits for the same record stream — it is the codec, not an estimate.

### 1. What compression means here

Each monitor record is a 128-bit packet + 64-bit timestamp. A raw (Tier-0) escape costs **3 × 64-bit beats**. The codec removes that two ways:

1. **Tier-1 templating** (64-bit slot): a CAM-hit record collapses to **1 beat**.  
Structural ceiling  $1 - 1/3 = 66.7\%$ .
2. **Half-beat packing** (32-bit slot): two compressible records share one beat (0.5 beat/record), raising the ceiling to  $1 - 0.5/3 = 83.3\%$ .

Realized reduction is dominated by the Tier-1 hit rate  $h$ , not slot size:

reduction =  $h \cdot (1 - b_t/3)$

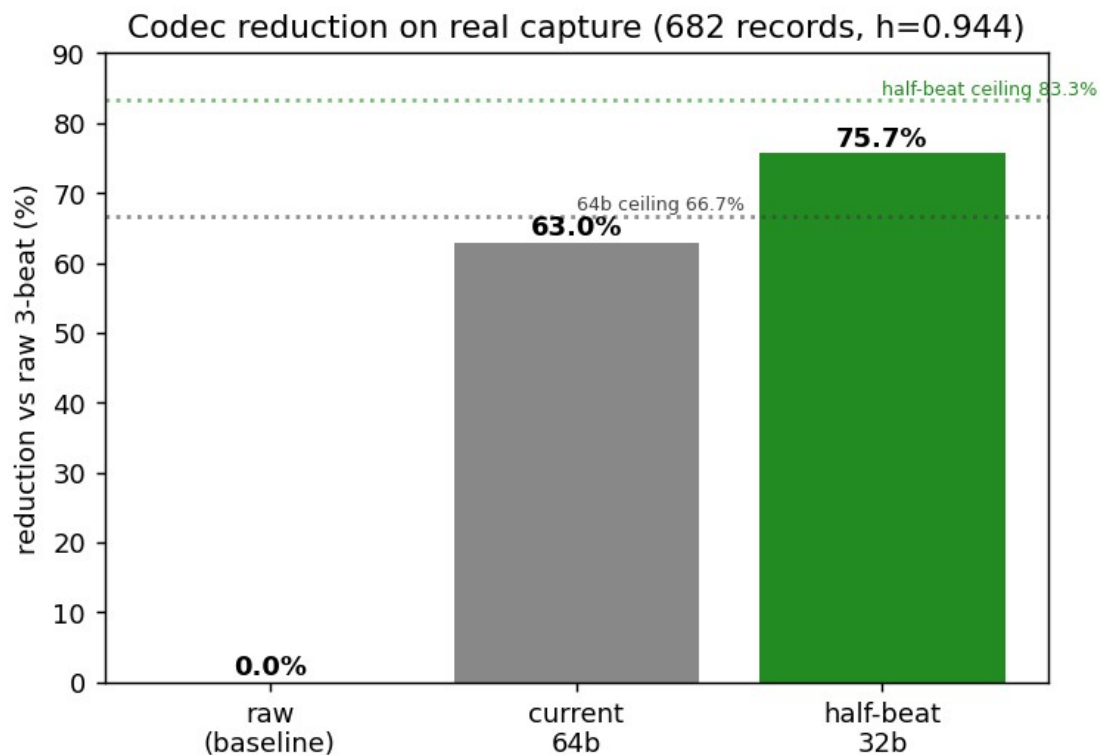
|                            |                 |                |
|----------------------------|-----------------|----------------|
| current ( $b_t = 1.0$ ):   | $h \cdot 0.667$ | ceiling 66.7 % |
| half-beat ( $b_t = 0.5$ ): | $h \cdot 0.833$ | ceiling 83.3 % |

So 80 % needs  $h \approx 0.96$ . The point of measuring on real traffic is to find the actual  $h$  and the fraction of hits that fit a 32-bit half-slot.

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## 2. Headline (real capture, 682 records, $h = 0.944$ )

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### codec comparison

| codec                   | beats      | reduction     |
|-------------------------|------------|---------------|
| raw baseline            | 2046       | —             |
| current 64-bit          | 758        | 63.0 %        |
| <b>half-beat 32-bit</b> | <b>498</b> | <b>75.7 %</b> |

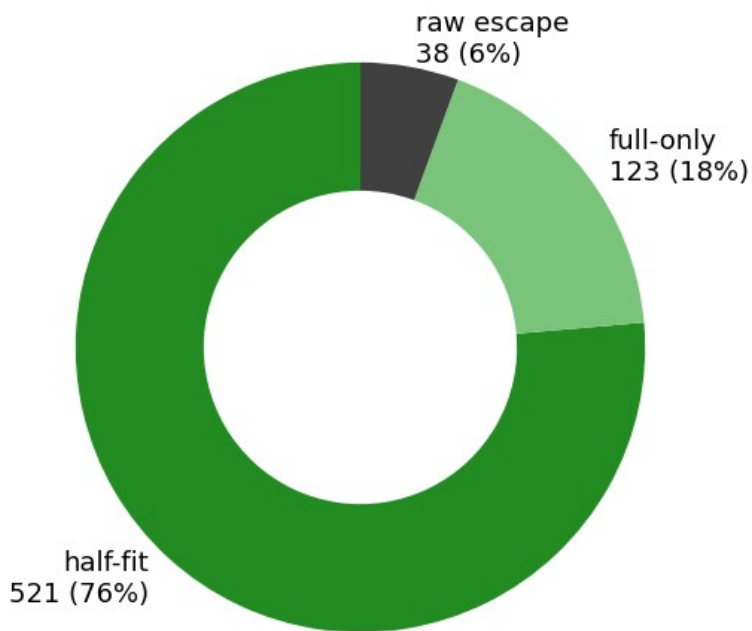
Half-beat packing turns the 63.0 % codec into 75.7 % on this trace — a **12.7-percentage-point** gain, **260 fewer beats** drained over UART for identical monitor coverage.

Why 75.7 % and not the 83.3 % ceiling:  $h = 0.944$  caps half-beat at  $0.944 \times 0.833 = 78.6 \%$ , and of the 644 hits, 123 fit only a full 64-bit beat (event\_data or  $\Delta t$ s too wide for a 23-bit half-payload), plus lone-half round-up. The gap to ceiling is real-traffic structure, not codec overhead.

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### 3. Per-record classification

Per-record classification — 682 records



#### *tier composition*

| class                                    | records | share | cost     |
|------------------------------------------|---------|-------|----------|
| half-fit (Tier-1, 32-bit slot)           | 521     | 76 %  | 0.5 beat |
| full-only (Tier-1, 64-bit slot)          | 123     | 18 %  | 1 beat   |
| raw escape (Tier-0, CAM miss / overflow) | 38      | 6 %   | 3 beats  |

Three quarters of records pack into half-slots. The 18 % full-only records are Tier-1 hits whose `event_data/dts` exceed the 23-bit half-payload; the 6 % raw escapes are first-occurrence CAM misses (one per distinct template) plus a handful of `dts` overflows.

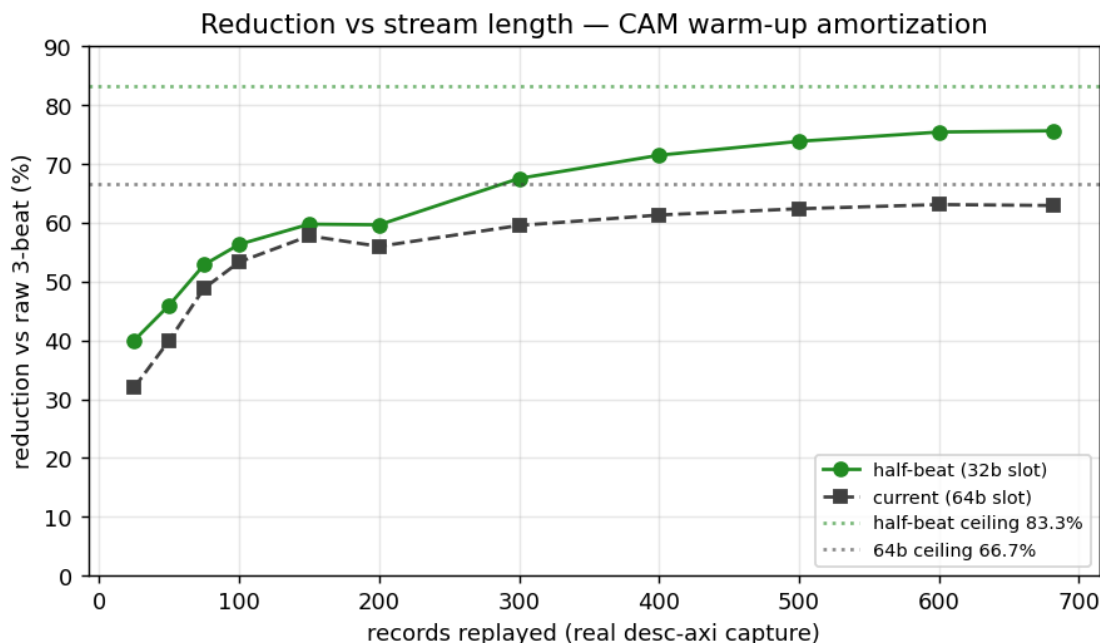
## 4. Half-beat wire format

A half-pair beat is {tag[63:60]=0x4, slotA[59:30], slotB[29:0]} — two 30-bit slots, each sub[1:0] + idx[4:0] + 23-bit payload:

```
HALF-A (absolute):  sub=1 | idx[4:0] | delta_ts[9:0] |  
event_data[12:0]  
HALF-C (delta):    sub=2 | idx[4:0] | delta_ts[9:0] | ed_delta[11:0]  
signed  
NOP (lone pad):    sub=0
```

$\text{delta\_ts} < 1024 \text{ cyc}$  ( $\sim 10.2 \mu\text{s}$  @ 100 MHz) for either form; HALF-A carries  $\text{event\_data} < 2^{13}$ , HALF-C a signed  $\text{ed\_delta} | \cdot | < 2^{12}$  for counters/sequential addresses. Tier-1 records that fit neither stay a full 64-bit beat; a lone trailing half rounds up to a full beat. Widths locked in `host/monbus_halfbeat_model.py` (lines 50–62), mirrored bit-exact by `rtl/amba/shared/monbus_halfbeat_packer.sv`.

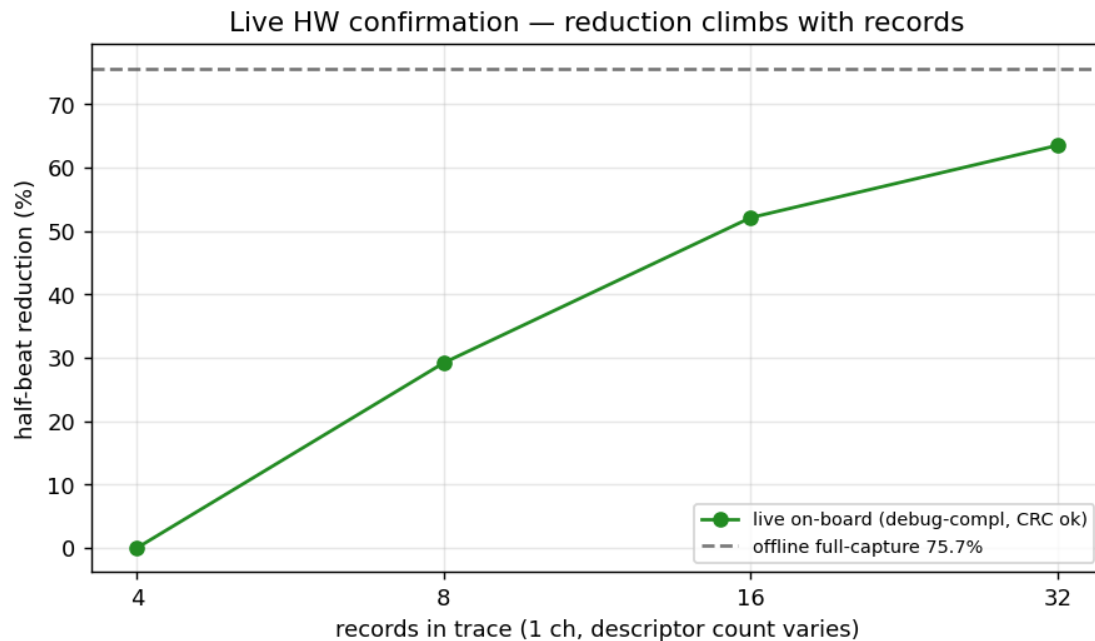
## 5. Reduction vs stream length — CAM warm-up



### *reduction vs length*

Replaying prefixes of the real capture shows the dominant real-world effect: **warm-up amortization**. Each distinct template costs one 3-beat raw escape on first sight; until those are paid off they dominate a short stream. As records accumulate the escapes amortize and reduction climbs toward the steady 75.7 %. This is why short captures under-report compression and the full-stream number is the representative one.

## 6. Live on-board confirmation



### *live confirmation*

The same codec is synthesized into the bitstream. `hb_measure.py` runs the compressed workload, drains the on-chip trace SRAM, and decodes the FPGA-emitted slots with the golden decoder — every clean point below round-trips bit-exact (decode=0K), confirming the HW slot stream is correct. 1-channel descriptor sweep (debug-compl monitor preset):

| records | half-beat beats | reduction                            |
|---------|-----------------|--------------------------------------|
| 4       | 12              | 0.0 % (all first-occurrence escapes) |
| 8       | 17              | 29.2 %                               |
| 16      | 23              | 52.1 %                               |
| 32      | 35              | 63.5 %                               |

The live curve tracks the offline warm-up curve (§5): reduction climbs with record count toward the full-capture 75.7 %.

### 6.1 Reproducibility — root-caused (was a reset bug, now fixed)

An earlier `hb_measure` run reported `8desc_8ch` as 261 beats one time and 405 another for the *identical* config. This was **proven** to be a measurement bug, not the codec:



- Localized on HW: 3 properly-reset runs give **bit-identical packets and identical per-template timestamp deltas** (only the absolute timer base shifts — it cancels in deltas), so the deterministic model yields the same compression every time.
- Root cause: hb\_measure fired only the soft-reset pulse, leaving per-channel + stat state stale between back-to-back invocations, which perturbed the captured event stream. **Fixed** (hb\_measure now does the full cluster reset: soft-reset → per-channel reset → clear-stats); with the fix, repeated 8desc\_8ch runs are bit-identical (212 rec / 405 beats / 36.3 %, CRC ok, every run).
- The compressor RTL is **bit-exact to the golden model in cosim on the exact captured stream** (val/amba/test\_monbus\_halfbeat.py, Phase 2 + DATASET\_OVERRIDE). The codec was never implicated.

## 6.2 OPEN — cluster state-accumulation wedge (not a clean-state bug)

**Correction.** An earlier draft here called this a reproducible 4–7-channel DMA hang / RTL bug. **That was wrong.** On a freshly reprogrammed board the full 1–8 channel debug - compl sweep passes (all CRC=ok), 8desc\_4ch passes 8/8 consecutive runs, and the hb\_measure path passes. The failures were an **accumulated-state wedge**.

After a long session of many mixed runs (wide resp-delay sweeps + the hb\_measure compression sweeps here), the cluster reaches a wedged state in which some configs hang (we saw 4–7 active channels under debug - compl time out at 120 s, zero beats, AXI\_RD\_COMPLETE=AXI\_WR\_COMPLETE=0). The per-run soft-reset / cluster-reset does **not** clear it — only a full FPGA reprogram does (make program). So it is a **reset-completeness / state-accumulation** gap in the harness, not a clean-state config or codec bug, and not data corruption. Tracked in rtl/amba/KNOWN\_ISSUES/axi\_monitor\_blockready\_hang\_partial\_channels.md. The compression numbers in §2–§6.1 are from clean-state runs and are unaffected.

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## 7. What we learned

1. **Half-beat is worth it on real traffic:** 63.0 % → 75.7 % (+12.7 pp) on the captured stream, for two 30-bit slots per beat.
2. **h dominates, not slot size.** At  $h = 0.944$  the half-beat ceiling is 78.6 %; we hit 75.7 % of the theoretical 83.3 %.

3. **Three quarters of records half-pack**; 18 % need a full beat (wide payload), 6 % escape (cold templates).
  4. **Warm-up is the real-world tax**. Short streams under-report; the cost is one 3-beat escape per distinct template, amortized over the run.
  5. **The HW codec is correct**. Live FPGA slots decode bit-exact and track the model; the codec was never the bug. The earlier non-reproducibility was a hb\_measure reset bug (fixed, §6.1).
  6. **Separate open issue (§6.2)**: a long mixed-run session can wedge the cluster so some configs hang — but a **freshly reprogrammed board passes the full sweep**, so it's a reset-completeness / state-accumulation gap in the harness, not a clean-state config, codec, or data-corruption bug. Cleared by make program.
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## Appendix: data files & reproduce

| File                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Contents                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| ../compression_dataset/<br>desc_axi_16desc_8ch_1MB.json                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | real 682-record on-board capture<br>(locked dataset)           |
| plots/*.png                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | figures above<br>(host/plot_compression_report.py)             |
| plots/summary.json                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | numeric summary (records, h, tier<br>split, beats, reductions) |
| <pre>cd flows-stream-bridge/host &amp;&amp; source \$REPO_ROOT/env_python # Authoritative, offline, bit-exact (no board needed): python3 plot_compression_report.py \     --capture ../reports/compression_dataset/desc_axi_16desc_8ch_1MB.json \     --outdir ../reports/compression/plots # Live on-board confirmation (keep workloads small; trace SRAM = 2048 beats; # reset the cluster between runs): python3 hb_measure.py --port /dev/ttyUSB2 --channels 1 \     --config 8desc_1ch_1MB --mon-config debug-compl</pre> |                                                                |
| <p>Codec spec: bin/TBClasses/monbus/monbus_compressor.py; RTL:<br/>rtl/amba/shared/monbus_{compressor,halfbeat_packer}.sv; cosim:<br/>val/amba/test_monbus_{compressor,halfbeat}.py.</p>                                                                                                                                                                                                                                                                                                                                       |                                                                |